2026 Summer Undergraduate Research Fellowship Interim Report I

Task-Driven Neural Geometries for Speech Production

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Background

Recent advancements in speech brain-computer interfaces (BCIs) by the Stanford Neural Prosthetics Translational Lab and associates [1–5] have successfully decoded phonemes from neural activity during attempted speech. Consequently, there has been additional interest in isolating and leveraging higher-level semantic features to improve decoding accuracy and robustness [6, 7]. Following previous work analyzing geometric structure of neural activity for both neuroscientific interest and potential BCI applications [8–20], we seek to understand both *how* and *why* the brain generates speech, converting high-level semantics into words, syllables, and phonemes. Ongoing neural data collection and analysis will answer the *how*. This project will answer the *why*—given emergent speech production patterns, what structural constraints in the brain elicited such geometry?

As in previous works [10, 14, 15, 17, 21–23], we use, as computational analogs to the biological brain, artificial task-driven models—specifically, recurrent neural networks (RNNs). With computational models, we can test how structural modifications to their systems alter their internal geometry. Thus, we can determine what physical constraints are responsible for the analogous biological behavior. By training RNNs to convert, for a dataset of sentences, language and vision-language model semantic embeddings into the appropriate phonemes, we aim to generate RNN hidden state geometries mirroring neural activity in biological speech production. Ultimately, we want to empirically identify how RNN architectural constraints affect the dynamical systems that consequently emerge internally—difficult to ascertain through theory alone.

Specifically, we want to experiment with the following RNN parameters, representing various potential biological constraints:

- Delay length between semantic input and accepting phoneme output
- Size and number of hidden states (memory size)
- Input and output format: simultaneous vs. serial, persistent vs. temporary
- Use of dropout and noise

Based on traditional theories [24–27] and results from recent studies [13, 17–19], we loosely pursue the following questions not fully answered in previous literature to characterize speech production:

- How does token context width scale with hidden state size?
- How many levels of speech (concepts, words, syllables, phonemes) are represented simultaneously?
- Does translation from words to phonemes occur concurrently for all tokens, or few tokens at once?
- When does such translation between different levels of speech occur?
- Prior to articulation, how early do the representations for each phoneme and their higher-level counterparts emerge?
- In hidden states, are token representations orthogonalized? To what extent do they overlap and how are they extricated?
- How do token representations encode expression order?

In the next section, we will discuss how we train RNN variants and analyze them to develop answers to these questions.

Progress

We have so far implemented most of the data collection, model training, and hidden state analysis pipelines. These will allow us to train our task-driven models and perform analysis to answer the questions of interest in later weeks.

Data Collection and Model Training

We collect 10,000 documents from the FineWeb dataset [28], deconstructing them into entries of individual sentences. For each sentence, we collect last-layer RoBERTa [29] and OpenCLIP [30] embeddings as high-level semantic representations; word tokens, decomposed via regex; and as low-level representations, syllable and phoneme tokens constructed from individual words using Syllabify [31], CMUdict [32], and g2p [33]. We then convert this dataset, for each training configuration, into the corresponding RNN input and output features for each timestep. Semantic embeddings are preserved in original dimensions, and word, syllable, and phoneme input tokens are converted to limited dimension embedding vectors. Phoneme output targets are one-hot encoded. When implementing a delay between initial input and first output, we add an extra silence logit to phoneme output with target 1 during the delay period and silence tokens between words. Furthermore, following [15], we add a go-cue dimension to the input features with value 1 when phoneme output begins to be expected.

So far, we have trained successful phoneme recall (serial phoneme input $\rightarrow$ delay $\rightarrow$ serial phoneme output) models on a cross-entropy loss objective. This has been used as a toy model for testing hidden state analysis implementations. We further implemented configurable input representations, delay lengths, hidden state size and depth, input and output formats, and input noise perturbations for model training in future experiments.

Analysis Modules

Having built a pipeline and dashboard for exporting hidden states from trained models, computing relevant metrics, and plotting them, we so far probe the hidden state geometry of our task-driven models in four ways:

SLIM. Using Superposition-based Linear Interpretable Models (SLIM), a technique pioneered in the group for finding the composition of phonemes in neural data, we can derive the projections of representation tokens at a given position and timestep in hidden state space. By using linear regression across validation sentence samples, we find weights $B_{i,t}$ corresponding to the phoneme at sequence position i, for the hidden state at timestep t, such that the hidden state at t

$$h_t = \sum_{i}^{N} B_{i,t} p_i + \epsilon,$$

where N is the sequence length and p_i is the one-hot encoding of the phoneme at position i. Then, $B_{i,t}\mathbf{e}_j$ provides the hidden state projection of the j-th phoneme in the vocabulary at time t, position i. Now, we can directly compare $B_{i,t}$ across i and t to determine:

1. The representation strength of phonemes at each timestep, by expression order (Fig. 1).
2. Orthogonality between phoneme representations at various positions and timesteps, via computing the cosine similarity between $B_{i_0,t_0}\mathbf{e}_j$ and $B_{i_1,t_1}\mathbf{e}_j$ averaged across all j (Fig. 2).
3. The dynamics of phoneme representations over time, visualized via principle component analysis (PCA).

We can further this technique to higher level representations such as syllables, words, and single-word semantic embeddings.

Linear Decoding. Given our interest in whether some representation level for a given phoneme is present in the hidden state at time t, we train linear decoders $L_{i,t}$ to predict the phoneme token at position i from h_t via Ridge regression. The validation accuracy of such decoders thus represents the "decodability" of token i from h_t (Fig. 3). Furthermore, just as we can compare hidden state representations with SLIM, we can see how well decoders trained for one timestep or position generalizes to other timesteps and positions.

Preparatory Activity in the Null Space. Previous works [8, 14, 19, 23] have shown that neural populations exhibit preparatory activations that do not immediately elicit but prime future activations for downstream motor activity (e.g., reaching). Preparatory activations occupy the output-null space of the final readout, as such having no effect on it for the current timestep. An equivalent analogy is that given a final layer readout matrix with weight $\mathbf{W}$ and hidden state $\mathbf{h}$, the component of $\mathbf{h}$ in the null space of $\mathbf{W}$, $\text{proj}_{\text{Ker}\,\mathbf{W}}\,\mathbf{h}$, has no effect on the final output $\mathbf{W}\mathbf{h} + \mathbf{b}$. We define the space orthogonal to the output-null space as the output-potent space. For our $\mathbf{W}$ analogy, the potent space would be the row space, Ran $\mathbf{W}^\intercal$.

As such, we expect our hidden states (and their population neural activity analogs) to lie primarily in the output-null space during the input and delay period and then transforming to occupy portions of the output-potent space. By examining the null and row spaces of the final readout layer of our RNN as described above, we can also analyze when the SLIM projections of each phoneme token projection navigates from the null to the potent space (Fig. 4).

As covered prior, we have an additional silence output logit. When computing the null and potent hidden state dimensions, this logit and its respective row in $\mathbf{W}$ are ignored. As nonlinear softmax activations were required for both deciphering

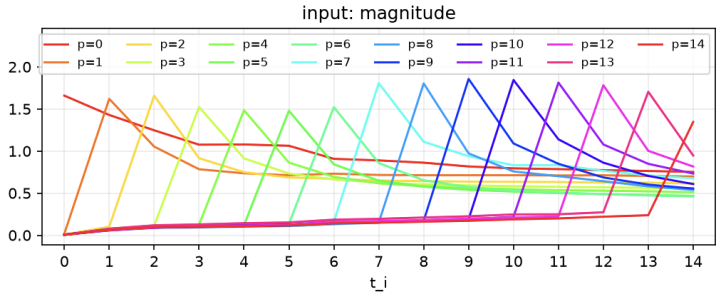


Figure 1: Sample plot of phoneme representation strengths during serial input for the phoneme recall task, computed via the average magnitude of SLIM weight $B_{i,t}$ columns. $p \equiv i$ is the phoneme's position in the sequence.

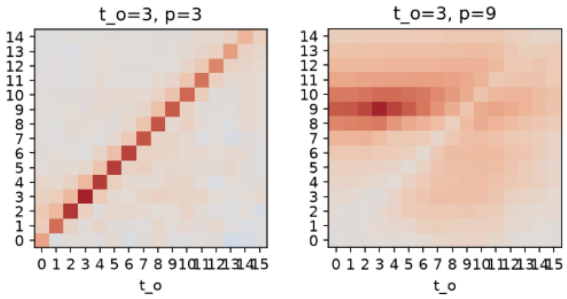


Figure 2: Sample plot of hidden state projection cosine similarities at various timesteps and positions during serial phoneme output for the phoneme recall task. Darker color indicates greater cosine similarity. **Left**: cosine similarities were measured between various $B_{i,t}$ and $B_{3,3}$; **Right**: between various $B_{i,t}$ and $B_{3,9}$. The strong diagonal for the $(t = 3, p = 3)$ anchor and other $(t = s, p = s)$ anchors indicate that phonemes temporally close to output are represented orthogonally to its projections in previous timesteps. Conversely, the horizontal smear for the $(t = 3, p = 3)$ anchor indicates that phonemes temporally far from output have nearly fixed representations across timesteps.

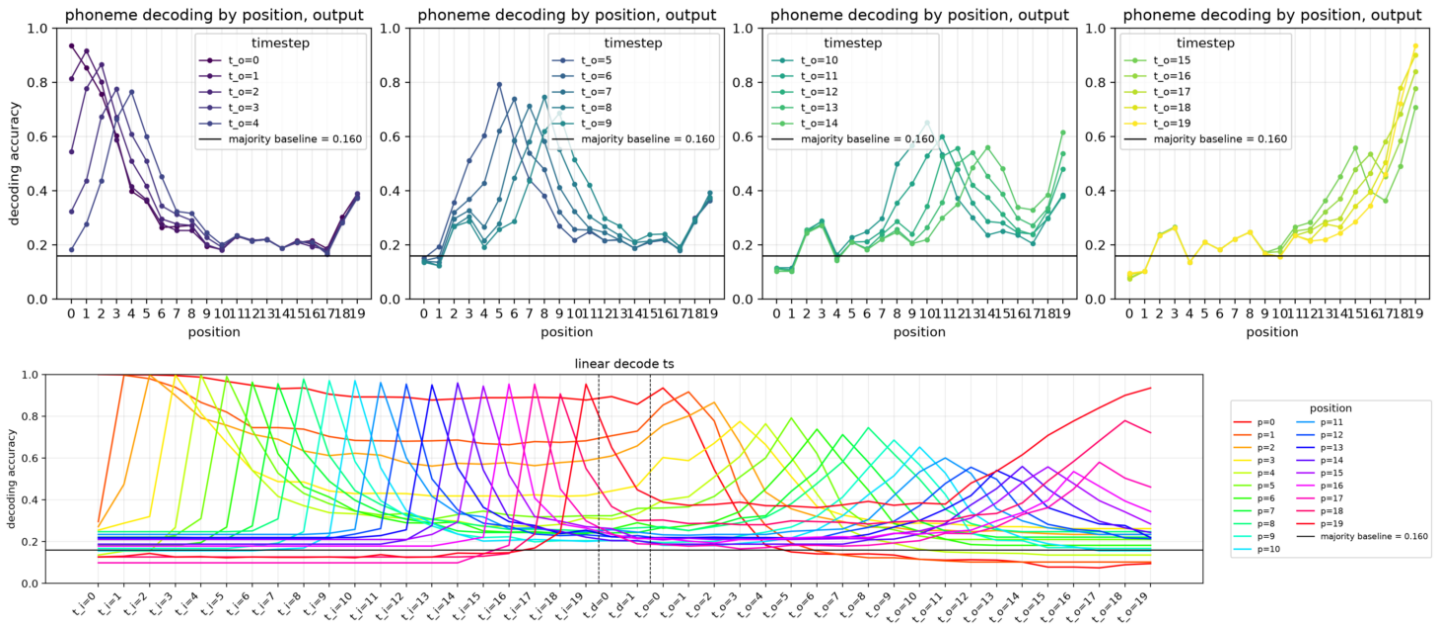


Figure 3: The accuracy of linear decoders $L_{i,t}$ on h_t in the validation set, as a measure of whether the phoneme is present and encoded in the RNN's internal representation for that timestep. **Top**: decoding accuracy of the phoneme at each position for given serial output timesteps, averaged across all validation hidden states. **Bottom**: mean decoding accuracy of phonemes across input, delay, and output timesteps for each position. In both plots, the decoding accuracy for a given position peaks during output at approximately the timestep on which the phoneme in that position is articulated.

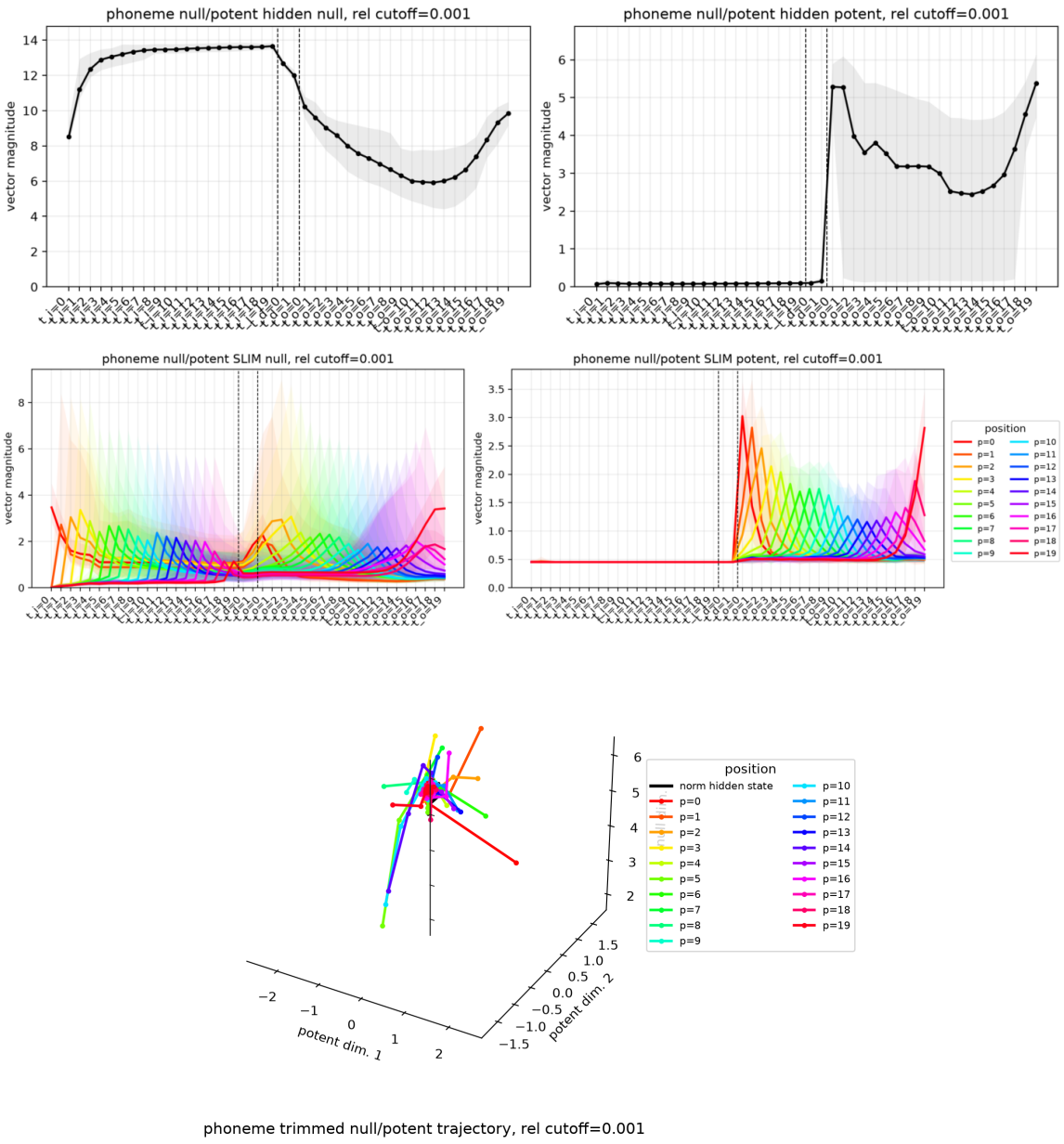


Figure 4: Null and potent spaces, and the magnitude of hidden states and SLIM projections in them. A mean-squared error (MSE) loss was used to force phoneme final layer output logits to 0 for silent tokens—a non-requisite otherwise given the softmax in the overall Cross-Entropy (CE) loss objective. The effect of this MSE loss is seen in the near-zero magnitudes in the output-potent space during the input and delay periods. Plots were generated from sequence length 20 samples for serial phoneme input and output. **Top**: mean hidden state magnitudes in the output-null (left) and output-potent (right) spaces across timesteps. Shaded areas represent middle 95% of values. **Middle**: mean SLIM phoneme hidden state projection magnitudes in the output-null (left) and output-potent (right) spaces across timesteps ($p \equiv i$). As expected, projections for each position extend furthest into output-potent space at their time of expression. **Bottom**: Trajectories of hidden state and SLIM projection vectors across time. For SLIM vectors, trajectories were truncated to timesteps between that phoneme's input and output. Hidden state vectors were normalized by sequence length. The two potent dimensions (horizontal) and one null dimension (vertical) were found via principle component analysis (PCA) dimension reduction of their respective output subspaces. As expected, the SLIM projection vectors begin exclusively along the null dimension and extend into the potent dimensions nearing output.

intended output during inference and encouraging convergence during training as part of the cross-entropy (CE) loss objective, this, if unmanaged, silences large deviations in final layer output. For example,

$$\text{softmax}([-1000, -10, 5, 3]) \approx [0, 2.69 \times 10^{-7}, 0.881, 0.119].$$

Effectively, phonemes with output of -1000 and -10 both read as inactive. This means that hidden states with significant output-potent components (with nonzero output -1000 or -10) have no effect on post-softmax activations, so our readout layer $\mathbf{W}$ *does not* have to learn to coerce silent phonemes to become 0 and its null and row spaces would not faithfully represent post-soft max outcomes. To overcome this, we add to the general CE loss an *efficiency* term, which punishes final layer readout values for deviating from 0 pre-softmax via mean-squared error (MSE) for silent token targets:

$$\mathcal{L} = \text{mean}_t \left[-\sum_{c \in \mathcal{C}} y_{t,c} \log \text{softmax}(z_t)_c \right] + \alpha \ \text{mean}_{t:y_{t,\langle sil \rangle}=1} \text{mean}_{c \in \mathcal{C} \setminus \{\langle sil \rangle\}} z_{t,c}^2$$

where $\mathcal{C}$ is the set of output classes, z_t are the readout logits ($\mathbf{W}\mathbf{h}_t + \mathbf{b}$), and α is the MSE loss weight.

Fixed Point Dynamics. Following [15, 34], we can treating RNN hidden state updates as a dynamical system. By locating bifurcations and fixed points—singular and topological attractors, saddle points, and unstable points—near hidden state trajectories, we can qualify the memory structures and computational processes with hidden states. For example, by identifying an unstable point near trajectories between hidden states with high encoding of word features and that of phonemic features, we can determine the exact timestep and condition under which word to phoneme translation occurs. Despite various attempts, we have yet to develop a satisfying procedure for locating fixed points and bifurcations with our current task structure. As such, implementing fixed points analysis will be an objective concurrent to obtaining results with the other analysis procedures.

Future Experiments

Subsequent work will first concentrate on completing and stress-testing analysis modules, before finalizing and initiating the training of full concept-to-phoneme translation RNN model variants. We anticipate that it may be difficult to train RNNs effective at converting high-dimensional language embeddings into phoneme output due to the computational complexity. As such, we would have to experiment with various RNN hidden layer configurations, and if unsuccessful, consider alternative neural architectures such as Transformers.

References

1. Willett, F. R. *et al.* A High-Performance Speech Neuroprosthesis. *Nature* **620,** 1031–1036 (2023/08/01, 2023).
2. Zhang, Y. *et al. Decoding Inner Speech with an End-to-End Brain-to-Text Neural Interface* Dec. 2025. arXiv: 2511.21740 [cs]. (2026).
3. Wairagkar, M. *et al.* An Instantaneous Voice-Synthesis Neuroprosthesis. *Nature* **644,** 145–152 (2025/08/01, 2025).
4. Yoon, S. *et al. Neural Decoding of Speech Using Deep Neural Ensembles* June 2026. (2026).
5. Fan, C. *et al. Plug-and-Play Stability for Intracortical Brain-Computer Interfaces: A One-Year Demonstration of Seamless Brain-to-Text Communication* in *Advances in Neural Information Processing Systems* **36** (Curran Associates, Inc., 2023), 42258–42270. (2026).
6. Zhang, Z. *et al.* Chisco: An EEG-based BCI Dataset for Decoding of Imagined Speech. *Scientific Data* **11,** 1265. ISSN: 2052-4463. (2026) (Nov. 2024).
7. Li, J., Singh, A. & Tandon, N. Transitioning from Speech to Language Brain–Computer Interfaces. *Nature Reviews Bioengineering*, 1–3. ISSN: 2731-6092. (2026) (May 2026).
8. Churchland, M. M. *et al.* Neural Population Dynamics during Reaching. *Nature* **487,** 51–56. ISSN: 0028-0836, 1476-4687. (2026) (July 2012).
9. Kaufman, M. T., Churchland, M. M., Ryu, S. I. & Shenoy, K. V. Cortical Activity in the Null Space: Permitting Preparation without Movement. *Nature Neuroscience* **17,** 440–448. ISSN: 1097-6256, 1546-1726. (2026) (Mar. 2014).
10. Russo, A. A. *et al.* Motor Cortex Embeds Muscle-like Commands in an Untangled Population Response. *Neuron* **97,** 953–966.e8. ISSN: 08966273. (2026) (Feb. 2018).
11. Stavisky, S. D. *et al.* Neural Ensemble Dynamics in Dorsal Motor Cortex during Speech in People with Paralysis. *eLife* **8,** e46015. ISSN: 2050-084X. (2026) (Dec. 2019).
12. Xie, Y. *et al.* Geometry of Sequence Working Memory in Macaque Prefrontal Cortex. *Science* **375,** 632–639. ISSN: 0036-8075, 1095-9203. (2026) (Feb. 2022).

13. Gwilliams, L., King, J.-R., Marantz, A. & Poeppel, D. Neural Dynamics of Phoneme Sequences Reveal Position-Invariant Code for Content and Order. *Nature Communications* **13,** 6606. ISSN: 2041-1723. (2026) (Nov. 2022).
14. Churchland, M. M. & Shenoy, K. V. Preparatory Activity and the Expansive Null-Space. *Nature reviews. Neuroscience* **25,** 213–236. ISSN: 1471-003X. (2026) (Apr. 2024).
15. Driscoll, L. N., Shenoy, K. & Sussillo, D. Flexible Multitask Computation in Recurrent Networks Utilizes Shared Dynamical Motifs. *Nature Neuroscience* **27,** 1349–1363. ISSN: 1546-1726. (2026) (July 2024).
16. Kunz, E. M. *et al.* Inner Speech in Motor Cortex and Implications for Speech Neuroprostheses. *Cell* **188,** 4658–4673.e17 (2025/08/21, 2025/12/28 2025).
17. Yang, B. & Yu, S. Orthogonal-Rotational Dynamics Supports Efficient Encoding and Updating for Streaming Information in Working Memory. *Journal of Neuroscience* **46.** ISSN: 0270-6474, 1529-2401. (2026) (Jan. 2026).
18. Gadonneix, J. *et al. Temporal Structure of the Language Hierarchy within Small Cortical Patches* Apr. 2026. arXiv: 2604.03021 [q-bio.NC]. (2026).
19. Krasa, B. A. *et al. Premotor Cortex Uses a Compositional Neural Geometry to Plan Words* Apr. 2026. (2026).
20. Deo, D. R. *et al.* A Mosaic of Whole-Body Representations on the Human Precentral Gyrus. *Nature,* 1–8. ISSN: 1476-4687. (2026) (June 2026).
21. Caucheteux, C. & King, J.-R. Brains and Algorithms Partially Converge in Natural Language Processing. *Communications Biology* **5,** 134. ISSN: 2399-3642. (2026) (Feb. 2022).
22. Marin Vargas, A. *et al.* Task-Driven Neural Network Models Predict Neural Dynamics of Proprioception. *Cell* **187,** 1745–1761.e19. ISSN: 0092-8674, 1097-4172. (2026) (Mar. 2024).
23. Schimel, M., Kao, T.-C. & Hennequin, G. When and Why Does Motor Preparation Arise in Recurrent Neural Network Models of Motor Control? *eLife* **12** (eds Diedrichsen, J. & de Lange, F. P.) RP89131. ISSN: 2050-084X. (2026) (Sept. 2024).
24. Levelt, W. J. M. *Speaking: From Intention to Articulation* (2026) (MIT press, 1993).
25. Levelt, W. J. M., Roelofs, A. & Meyer, A. S. A Theory of Lexical Access in Speech Production. *Behavioral and Brain Sciences* **22,** 1–38. ISSN: 1469-1825, 0140-525X. (2026) (Feb. 1999).
26. Dell, G. S. A Spreading-Activation Theory of Retrieval in Sentence Production. *Psychological review* **93,** 283. (2026) (1986).
27. Bohland, J. W., Bullock, D. & Guenther, F. H. Neural Representations and Mechanisms for the Performance of Simple Speech Sequences. *Journal of Cognitive Neuroscience* **22,** 1504–1529. ISSN: 0898-929X. (2026) (July 2010).
28. Penedo, G. *et al. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale* in *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track* (Nov. 2024). (2026).
29. Liu, Y. *et al. RoBERTa: A Robustly Optimized BERT Pretraining Approach* July 2019. arXiv: 1907.11692 [cs.CL]. (2026).
30. Cherti, M. *et al. Reproducible Scaling Laws for Contrastive Language-Image Learning* in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (June 2023), 2818–2829. arXiv: 2212.07143 [cs.LG]. (2026).
31. *Syllabify: Automatically Convert Plain Text into Phonemes (US English Pronunciation) and Syllabify* (2026).
32. *The CMU Pronouncing Dictionary* http://www.speech.cs.cmu.edu/cgi-bin/cmudict. (2026).
33. Park Kyubyong & Kim, J. *g2pE* 2019.
34. Sussillo, D. & Barak, O. Opening the Black Box: Low-Dimensional Dynamics in High-Dimensional Recurrent Neural Networks. *Neural Computation* **25,** 626–649. ISSN: 0899-7667. (2026) (Mar. 2013).